

HARMONIC ANALYSIS

PROBLEM SET



Fourier Analysis

Chapters I–IX

Large-Type Mathematical Study Workbook

PRINCETON / INSTITUTE FOR ADVANCED STUDY INSPIRED STUDY EDITION

Nine-chapter proof workbook with reconstructed statements, examples, exercises, problems, and generous proof space.



Fuld Hall – Institute for Advanced Study, Princeton, New Jersey.

Private study workbook – not an official Princeton / IAS publication.



Chapter Map

LARGE-TYPE WORKING ATLAS

Chapter I - Fourier Series and Integrals

Fourier coefficients, Dirichlet and Jordan convergence, Fejér and Poisson kernels, Fourier transform, Schwartz space, Plancherel, interpolation, and convergence of Fourier integrals.

Chapter II - The Hardy-Littlewood Maximal Function

Approximate identities, weak-type ideas, Marcinkiewicz interpolation, maximal estimates, dyadic decomposition, differentiation, weighted bounds, and further maximal operators.

Chapter III - The Hilbert Transform

Principal values, M. Riesz and Kolmogorov bounds, truncated transforms, Cotlar inequality, multipliers, and translation-invariant operators.



Chapter IV - Singular Integrals (I)

Homogeneous singular kernels, Fourier multipliers, method of rotations, rough maximal functions, odd kernels, Riesz transforms, and principal-value convergence.



CHAPTER I

Fourier Series and Integrals

Fourier coefficients, convergence criteria, summability, Fourier transform, and interpolation

The core statements run from Chapter 1, pp. 1–24, with later notes and further results on pp. 20–24.



EXERCISE E01 CORE

Fourier coefficients and orthogonality

SOURCE LOCUS: Chapter 1, §1, equations (1.2)–(1.5), pp. 1–2

CORE

For a 1-periodic $f \in L^1(\mathbb{T})$ define $\widehat{f}(k) = \int_0^1 f(x) e^{-2\pi i k x} dx$.

- (a) Prove the orthogonality relation for $e^{2\pi i k x}$.
- (b) Derive the coefficient formula from a uniformly convergent Fourier series.
- (c) Explain why the symmetric series is the natural formulation used later.



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Exercise E01

Fourier coefficients and orthogonality

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E01

Fourier coefficients and orthogonality

WORK PAGE 2





EXERCISE E02 CORE

The Dirichlet kernel

SOURCE LOCUS: Chapter 1, §2, equation (1.6), p. 3

CORE

Derive $D_N(t) = \sum_{k=-N}^N e^{2\pi i k t} = \frac{\sin(\pi(2N+1)t)}{\sin(\pi t)}$ and verify its normalization and pointwise size estimate away from the origin.

(a) Derive the geometric-series formula.

(b) Show $\int_{-1/2}^{1/2} D_N(t) dt = 1$.

(c) Prove $|D_N(t)| \lesssim |\sin(\pi t)|^{-1}$.



PRINCETON PROOF SHEET

Exercise E02

The Dirichlet kernel

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E02

The Dirichlet kernel

WORK PAGE 2



**EXERCISE E03 ADVANCED****Dini's criterion**

SOURCE LOCUS: Chapter 1, Theorem 1.1, pp. 2–4

ADVANCED

Assume that for some x and $\delta > 0$, $\int_{|t|<\delta} \frac{|f(x+t)-f(x)|}{|t|} dt < \infty$.
Reconstruct the proof that $S_N f(x) \rightarrow f(x)$.

- (a) Insert the Dirichlet kernel representation.
- (b) Use $\int D_N = 1$ to isolate $f(x)$.
- (c) Split into local and non-local pieces.
- (d) Apply the Riemann-Lebesgue lemma to the integrable error term.



PRINCETON PROOF SHEET

Exercise E03

Dini’s criterion

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E03

Dini’s criterion

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E03

Dini’s criterion

WORK PAGE 3



**EXERCISE E04 CORE****Jordan's criterion**

SOURCE LOCUS: Chapter 1, Theorem 1.2, pp. 3–5

CORE

Suppose f has bounded variation near x . Show that $S_N f(x) \rightarrow \frac{1}{2}(f(x+) + f(x-))$. Reduce to the monotone case and use the symmetry of the Dirichlet kernel.



PRINCETON PROOF SHEET

Exercise E04

Jordan’s criterion

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E04

Jordan’s criterion

WORK PAGE 2



**PROBLEM P01 CHALLENGE****Riemann localization from Riemann-Lebesgue****CHALLENGE**

SOURCE LOCUS: Chapter 1, Theorem 1.3 and Lemma 1.4, pp. 3–4

Assume f vanishes on a neighborhood of x . Reconstruct the localization proof by writing $S_N f(x)$ as oscillatory integrals whose amplitudes are integrable.



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Problem P01

Riemann localization

WORK PAGE 1





PRINCETON PROOF SHEET

Problem P01

Riemann localization

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P01

Riemann localization

WORK PAGE 3



**EXAMPLE X01 CORE****Fejér kernel and Cesàro summability**

SOURCE LOCUS: Chapter 1, §5, pp. 8–10

CORE

Let $F_N(t) = \frac{1}{N+1} \left(\frac{\sin(\pi(N+1)t)}{\sin(\pi t)} \right)^2$. Prove positivity, unit integral, and concentration at 0.

- (a) Derive $F_N = \frac{1}{N+1} \sum_{k=0}^N D_k$.
- (b) Prove $F_N \geq 0$ and $\|F_N\|_1 = 1$.
- (c) Deduce norm convergence of Cesàro means for $1 < p < \infty$ and for continuous periodic functions.



PRINCETON PROOF SHEET

Example X01

Fej' er kernel and Ces` aro summability

WORK PAGE 1





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Example X01

Fejér kernel and Cesàro summability

WORK PAGE 2





PRINCETON PROOF SHEET

Example X01

Fejér kernel and Cesàro summability

WORK PAGE 3



**EXAMPLE X02 ADVANCED****Poisson summability and the Dirichlet problem****ADVANCED**

SOURCE LOCUS: Chapter 1, §5, pp. 10–11 and §9, pp. 17–19

Show that the Poisson kernel is positive, has unit integral, and yields $P_r * f \rightarrow f$ in L^p for $1 < p < \infty$ and uniformly for continuous f . Interpret the harmonic extension as a Dirichlet solution.



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Example X02

Poisson summability and the Dirichlet problem

WORK PAGE 1





PRINCETON PROOF SHEET

Example X02

Poisson summability and the Dirichlet problem

WORK PAGE 2





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Example X02

Poisson summability and the Dirichlet problem

WORK PAGE 3



**EXERCISE E05 CORE****The Gaussian as a Fourier fixed point**

SOURCE LOCUS: Chapter 1, Lemma 1.14, p. 13

CORE

Let $f(x) = e^{-\pi|x|^2}$. Reconstruct the differential-equation argument proving $\widehat{f} = f$.

- (a) Use $f' + 2\pi x f = 0$ in one dimension.
- (b) Fourier transform the equation and identify the initial value.
- (c) Extend to $\mathbb{R}^n$ by product structure.



PRINCETON PROOF SHEET

Exercise E05

Gaussian Fourier fixed point

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E05

Gaussian Fourier fixed point

WORK PAGE 2



**PROBLEM P02 ADVANCED****Plancherel and inversion on L^2**

SOURCE LOCUS: Chapter 1, Theorem 1.18, pp. 15–16

ADVANCED

Use density of $\mathcal{S}$ and the identity $\int f \overline{g} = \int \widehat{f} \overline{\widehat{g}}$ to extend the Fourier transform to an isometry on L^2 . Then prove the L^2 inversion statements by truncation.



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Problem P02

Plancherel and inversion

WORK PAGE 1





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Problem P02

Plancherel and inversion

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P02

Plancherel and inversion

WORK PAGE 3



**EXERCISE E06 CORE****Hausdorff–Young and Young’s inequality**

SOURCE LOCUS: Chapter 1, Corollaries 1.20–1.21, pp. 16–17

CORE

Reconstruct how Riesz–Thorin gives Hausdorff–Young and then Young’s convolution inequality. Make the interpolation exponents explicit.



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Exercise E06

Hausdorff-Young and Young

WORK PAGE 1





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Exercise E06

Hausdorff-Young and Young

WORK PAGE 2



**PROBLEM P03 CHALLENGE****Gibbs phenomenon and sharp overshoot**

SOURCE LOCUS: Chapter 1, §10.4, p. 21

CHALLENGE

For $f = \text{sgn}$ on $(-1/2, 1/2)$, explain why partial sums converge pointwise yet overshoot persists. Reconstruct the limiting overshoot integral given in the notes.



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Problem P03

Gibbs phenomenon

WORK PAGE 1





PRINCETON PROOF SHEET

Problem P03

Gibbs phenomenon

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P03

Gibbs phenomenon

WORK PAGE 3





CHAPTER II

The Hardy–Littlewood Maximal Function

Approximate identities, weak type estimates, interpolation, differentiation, and weights

The core statements run from Chapter 2, pp. 25–38; the notes continue through pp. 39–48.

**EXERCISE E07 CORE****Approximation of the identity in L^p**

SOURCE LOCUS: Chapter 2, Theorem 2.1, pp. 25–26

CORE

Let $\phi_t(x) = t^{-n}\phi(t^{-1}x)$ with $\int \phi = 1$. Reconstruct the proof that $\phi_t * f \rightarrow f$ in L^p for $1 < p < \infty$ and uniformly for $f \in C_0$.



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Exercise E07

Approximation of the identity

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E07

Approximation of the identity

WORK PAGE 2



**EXERCISE E08 CORE****Closedness from a weak maximal estimate**

SOURCE LOCUS: Chapter 2, Theorem 2.2, pp. 27–28

CORE

Let $T_t f \rightarrow f$ for a dense class and suppose $T^* f = \sup_t |T_t f|$ is weak (p, p) . Show that the set of functions for which $T_t f \rightarrow f$ a.e. is closed in L^p .



PRINCETON PROOF SHEET

Exercise E08

Closedness principle

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E08

Closedness principle

WORK PAGE 2



**EXERCISE E09 ADVANCED****Marcinkiewicz interpolation**

SOURCE LOCUS: Chapter 2, Theorem 2.4, pp. 28–30

ADVANCED

Reconstruct the decomposition $f = f_0 + f_1$ at height $c\lambda$ and derive a strong (p, p) estimate from weak endpoint estimates. Keep track of the distribution-function formula.



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Exercise E09

Marcinkiewicz interpolation

WORK PAGE 1





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Exercise E09

Marcinkiewicz interpolation

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E09

Marcinkiewicz interpolation

WORK PAGE 3



**EXERCISE E10 CORE****One-dimensional Hardy–Littlewood weak $(1, 1)$**

SOURCE LOCUS: Chapter 2, Theorem 2.5, pp. 30–31

CORE

Using the interval covering lemma, prove $|\{Mf > \lambda\}| \leq C\lambda^{-1}\|f\|_1$ in one dimension, then recover strong (p, p) by interpolation.



PRINCETON PROOF SHEET

Exercise E10

One-dimensional maximal weak type

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E10

One-dimensional maximal weak type

WORK PAGE 2



**PROBLEM P04 ADVANCED****Dyadic maximal function and differentiation**

SOURCE LOCUS: Chapter 2, §5, Theorem 2.10, pp. 32–35

ADVANCED

Reconstruct the dyadic decomposition $\{M_d f > \lambda\} = \bigcup Q_j$ and prove weak $(1, 1)$. Then establish $E_k f \rightarrow f$ a.e. for $f \in L^1_{\text{loc}}$.



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Problem P04

Dyadic maximal function

WORK PAGE 1





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Problem P04

Dyadic maximal function

WORK PAGE 2





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Problem P04

Dyadic maximal function

WORK PAGE 3



**EXERCISE E11 CORE****Lebesgue differentiation theorem**

SOURCE LOCUS: Chapter 2, Corollary 2.13, p. 36

CORE

Deduce differentiation of locally integrable functions from the weak estimate for M and Theorem 2.2. Include the sharper mean-oscillation limit.



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Exercise E11

Lebesgue differentiation

WORK PAGE 1





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Exercise E11

Lebesgue differentiation

WORK PAGE 2



**EXAMPLE X03 ADVANCED****The L^1 obstruction for the maximal function**

SOURCE LOCUS: Chapter 2, Proposition 2.14, p. 36

ADVANCED

Show that if $f \in L^1$ is nonzero then $Mf \notin L^1$. Use a positive mass on a fixed ball to get the tail lower bound $Mf(x) \gtrsim |x|^{-n}$.



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Example X03

Failure of strong (1, 1)

WORK PAGE 1





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Example X03

Failure of strong (1, 1)

WORK PAGE 2





PRINCETON PROOF SHEET

Example X03

Failure of strong (1, 1)

WORK PAGE 3



**EXERCISE E12 ADVANCED** **$L \log L$ control**

SOURCE LOCUS: Chapter 2, Theorem 2.15 and Theorem 2.18, pp. 37–42

ADVANCED

Reconstruct the local integrability estimate for Mf on a bounded set and explain why $f \log^+ f \in L^1$ is the natural threshold.



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Exercise E12

Llog Lcontrol

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E12

Llog Lcontrol

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E12

Llog Lcontrol

WORK PAGE 3



**PROBLEM P05 CHALLENGE****Covering lemmas and non-doubling measures**

SOURCE LOCUS: Chapter 2, Theorems 2.19–2.20, pp. 43–45

CHALLENGE

Compare the Vitali-type and Besicovitch–Morse coverings. Explain why the non-centered maximal theorem for general measures needs a doubling hypothesis in dimensions $n > 1$.



PRINCETON PROOF SHEET

Problem P05

Covering lemmas and measures

WORK PAGE 1





PRINCETON PROOF SHEET

Problem P05

Covering lemmas and measures

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P05

Covering lemmas and measures

WORK PAGE 3



**EXERCISE E13 CHALLENGE****Strong maximal function**

SOURCE LOCUS: Chapter 2, §8.8, pp. 45–46

CHALLENGE

Show the L^p boundedness of the strong maximal function by iterating one-dimensional maximal operators, and contrast it with its sharp $L \log^{n-1} L$ endpoint behavior.



PRINCETON PROOF SHEET

Exercise E13

Strong maximal function

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E13

Strong maximal function

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E13

Strong maximal function

WORK PAGE 3





CHAPTER III

The Hilbert Transform

Principal values, M. Riesz–Kolmogorov bounds, truncations, and multiplier structure

The core statements run from Chapter 3, pp. 49–61, followed by notes through p. 68.

**EXERCISE E14 CORE****Conjugate Poisson kernel**

SOURCE LOCUS: Chapter 3, §1, pp. 49–50

CORE

Derive the harmonic conjugate $v = Q_t * f$ of $u = P_t * f$ and identify $P_t + iQ_t$ as the boundary value of an analytic kernel.



PRINCETON PROOF SHEET

Exercise E14

Conjugate Poisson kernel

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E14

Conjugate Poisson kernel

WORK PAGE 2



**EXERCISE E15 CORE****Principal value of $1/x$**

SOURCE LOCUS: Chapter 3, Proposition 3.1, p. 50

CORE

Show that $\text{p.v.}(1/x)$ defines a tempered distribution and that $Q_t \rightarrow \pi^{-1} \text{p.v.}(1/x)$ in $\mathcal{S}'$.



PRINCETON PROOF SHEET

Exercise E15

Principal value of $1/x$

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E15

Principal value of $1/x$

WORK PAGE 2



**EXERCISE E16 ADVANCED****The Hilbert transform on L^2**

SOURCE LOCUS: Chapter 3, equations (3.4)–(3.6), pp. 50–52

ADVANCED

Using the multiplier $-i \operatorname{sgn} \xi$, prove $\|Hf\|_2 = \|f\|_2$, $H^2 = -I$, and the skew-adjoint identity.



PRINCETON PROOF SHEET

Exercise E16

Hilbert transform on L^2

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E16

Hilbert transform on L^2

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E16

Hilbert transform on L^2

WORK PAGE 3



**PROBLEM P06 ADVANCED****Kolmogorov weak $(1, 1)$ estimate**

SOURCE LOCUS: Chapter 3, Theorem 3.2, pp. 51–54

ADVANCED

Reconstruct the Calderón–Zygmund decomposition $f = g + b$ and estimate the good and bad pieces outside the doubled intervals.



PRINCETON PROOF SHEET

Problem P06

Kolmogorov weak type (1, 1)

WORK PAGE 1





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Problem P06

Kolmogorov weak type (1, 1)

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P06

Kolmogorov weak type (1, 1)

WORK PAGE 3



**EXERCISE E17 CORE****M. Riesz theorem by interpolation and duality**

SOURCE LOCUS: Chapter 3, Theorem 3.2, pp. 53–54

CORE

Starting from weak $(1, 1)$ and strong $(2, 2)$, derive strong (p, p) for $1 < p < 2$, then use duality for $p > 2$.



PRINCETON PROOF SHEET

Exercise E17

M. Riesz theorem

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E17

M. Riesz theorem

WORK PAGE 2



**EXERCISE E18 ADVANCED****Cotlar’s inequality for truncated transforms**

SOURCE LOCUS: Chapter 3, Lemma 3.5, pp. 56–58

ADVANCED

Show that $H^* f(x) \lesssim M(Hf)(x) + CMf(x)$ by comparing the truncated kernel with a smooth approximation of p.v. $(1/x)$.



PRINCETON PROOF SHEET

Exercise E18

Cotlar’s inequality

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E18

Cotlar’s inequality

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E18

Cotlar’s inequality

WORK PAGE 3



**EXERCISE E19 CORE****Interval multipliers**

SOURCE LOCUS: Chapter 3, Proposition 3.6, pp. 58–59

CORE

Express $S_{a,b}$ through modulated Hilbert transforms and deduce its uniform L^p bound.



PRINCETON PROOF SHEET

Exercise E19

Interval multipliers

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E19

Interval multipliers

WORK PAGE 2



**EXAMPLE X04 ADVANCED****Bounded-variation multipliers**

SOURCE LOCUS: Chapter 3, Corollary 3.8, pp. 59–60

ADVANCED

Let m have finite total variation. Reconstruct the Stieltjes representation of m and use Minkowski to prove that m is an L^p multiplier.



PRINCETON PROOF SHEET

Example X04

Bounded-variation multipliers

WORK PAGE 1





PRINCETON PROOF SHEET

Example X04

Bounded-variation multipliers

WORK PAGE 2





PRINCETON PROOF SHEET

Example X04

Bounded-variation multipliers

WORK PAGE 3



**PROBLEM P07 CHALLENGE****Translation-invariant operators**

SOURCE LOCUS: Chapter 3, Theorem 3.12 and Proposition 3.13, pp. 66–67

CHALLENGE

Reconstruct the translation argument showing that a nonzero translation-invariant operator bounded $L^p \rightarrow L^q$ must have $p \leq q$, and relate L^2 boundedness to an L^∞ multiplier.



PRINCETON PROOF SHEET

Problem P07

Translation invariance and multipliers

WORK PAGE 1





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Problem P07

Translation invariance and multipliers

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P07

Translation invariance and multipliers

WORK PAGE 3



**EXAMPLE X05 CORE****Hilbert transform of a characteristic function**

SOURCE LOCUS: Chapter 3, §3 and §6.9, pp. 54 and 68

CORE

Compute the Hilbert transform of $\chi_{(0,1)}$ and explain directly why the strong $(1, 1)$ and strong (∞, ∞) bounds fail.



PRINCETON PROOF SHEET

Example X05

Hilbert transform of an interval

WORK PAGE 1





PRINCETON PROOF SHEET

Example X05

Hilbert transform of an interval

WORK PAGE 2





CHAPTER IV

Singular Integrals (I)

Homogeneous kernels, method of rotations, rough maximal operators, odd kernels, and Riesz transforms

The core statements run from Chapter 4, pp. 69–84; further notes and results continue through p. 90.

**EXERCISE E20 CORE****Cancellation is necessary**

SOURCE LOCUS: Chapter 4, Proposition 4.1, p. 69–70

CORE

For Tf defined from $\Omega(y')/|y|^n$, show that existence of the principal-value limit forces $\int_{S^{n-1}} \Omega \, d\sigma = 0$.



PRINCETON PROOF SHEET

Exercise E20

Necessary cancellation

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E20

Necessary cancellation

WORK PAGE 2





EXERCISE E21 ADVANCED

Homogeneous distributions and Fourier transforms**ADVANCED**

SOURCE LOCUS: Chapter 4, Proposition 4.3, pp. 70–71

Prove: if a tempered distribution is homogeneous of degree α , its Fourier transform is homogeneous of degree $-n - \alpha$. Apply this to $|x|^{-\alpha}$.



PRINCETON PROOF SHEET

Exercise E21

Homogeneous distributions

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E21

Homogeneous distributions

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E21

Homogeneous distributions

WORK PAGE 3



**PROBLEM P08 CHALLENGE****Fourier transform of rough homogeneous kernels****CHALLENGE**

SOURCE LOCUS: Chapter 4, Theorem 4.4, pp. 71–73

Reconstruct the derivation of the multiplier formula involving the logarithmic term and the sign term. Make the role of the zero-average condition explicit.



PRINCETON PROOF SHEET

Problem P08

Fourier transform of rough kernels

WORK PAGE 1





PRINCETON PROOF SHEET

Problem P08

Fourier transform of rough kernels

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P08

Fourier transform of rough kernels

WORK PAGE 3



**EXAMPLE X06 CORE****Newtonian and logarithmic potentials**

SOURCE LOCUS: Chapter 4, §1, pp. 69–70

CORE

Identify the singular integral kernels arising from the first derivatives of the Newtonian potential and the mixed second derivative of the logarithmic potential in $\mathbb{R}^2$.



PRINCETON PROOF SHEET

Example X06

Newtonian and logarithmic potentials

WORK PAGE 1





PRINCETON PROOF SHEET

Example X06

Newtonian and logarithmic potentials

WORK PAGE 2



**EXERCISE E22 CORE****Directional operators**

SOURCE LOCUS: Chapter 4, Proposition 4.6, pp. 73–74

CORE

Given a one-dimensional L^p -bounded operator T , define the directional operators T_u and prove their uniform $L^p(\mathbb{R}^n)$ bound by Fubini.



PRINCETON PROOF SHEET

Exercise E22

Directional operators

WORK PAGE 1





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Exercise E22

Directional operators

WORK PAGE 2



**EXERCISE E23 ADVANCED****Method of rotations for rough maximal functions**

SOURCE LOCUS: Chapter 4, Corollary 4.7, pp. 74–75

ADVANCED

Use polar coordinates to dominate the rough maximal operator by an average of directional maximal functions, and deduce L^p boundedness for $p > 1$.



PRINCETON PROOF SHEET

Exercise E23

Method of rotations

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E23

Method of rotations

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E23

Method of rotations

WORK PAGE 3



**EXERCISE E24 ADVANCED****Odd kernels and the Hilbert transform**

SOURCE LOCUS: Chapter 4, Corollary 4.8, pp. 75–76

ADVANCED

Assume Ω is odd and integrable on S^{n-1} . Rewrite the principal-value operator as an average of directional Hilbert transforms and deduce its L^p boundedness.



PRINCETON PROOF SHEET

Exercise E24

Odd-kernel singular integrals

WORK PAGE 1





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Exercise E24

Odd-kernel singular integrals

WORK PAGE 2





PRINCETON PROOF SHEET

Exercise E24

Odd-kernel singular integrals

WORK PAGE 3



**EXERCISE E25 CORE****Maximal truncations**

SOURCE LOCUS: Chapter 4, Corollary 4.9, pp. 76–77

CORE

From the directional maximal Hilbert transform bound, derive the strong (p, p) bound for the maximal truncation T^* . Explain why this yields a.e. existence of the principal value.



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Exercise E25

Maximal truncations

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E25

Maximal truncations

WORK PAGE 2



**EXAMPLE X07 ADVANCED****Riesz transforms**

SOURCE LOCUS: Chapter 4, equations (4.7)–(4.9), pp. 76–77

ADVANCED

Verify the normalization of the Riesz kernels and derive $\sum_{j=1}^n R_j^2 = -I$ on L^p .



PRINCETON PROOF SHEET

Example X07

Riesz transforms

WORK PAGE 1





PRINCETON PROOF SHEET

Example X07

Riesz transforms

WORK PAGE 2





PRINCETON PROOF SHEET

Example X07

Riesz transforms

WORK PAGE 3



**PROBLEM P09 CHALLENGE****Even kernels and the method of rotations**

SOURCE LOCUS: Chapter 4, §§4–5, pp. 77–83

CHALLENGE

Trace the chapter’s even-kernel route: identify why the odd-kernel argument no longer applies directly, and reconstruct the multiplier-side strategy using the one-dimensional building blocks and the operator algebra.



PRINCETON PROOF SHEET

Problem P09

Even kernels and operator algebra

WORK PAGE 1





PRINCETON PROOF SHEET

Problem P09

Even kernels and operator algebra

WORK PAGE 2





PRINCETON PROOF SHEET

Problem P09

Even kernels and operator algebra

WORK PAGE 3





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Course _____

Date _____

This Chapter V workbook follows the supplied text section by section. It uses 22 pt body type, separate proof sheets, two full answer pages for core exercises, and three for advanced/challenge items. Statements are reconstructed in LaTeX from the chapter; no textbook-page screenshots are used.

PRIMARY SOURCE

Javier Duoandikoetxea, *Fourier Analysis*, translated and revised by David Cruz-Uribe, SFO, Graduate Studies in Mathematics 29, American Mathematical Society



CHAPTER MAP

Large-type working atlas

§1 The Calderón-Zygmund theorem

Hörmander smoothness, weak $(1, 1)$, interpolation, and Dini-type angular conditions.

§2 Truncations and principal values

Uniform Fourier bounds for truncated kernels, principal-value existence, and the oscillatory kernel $|x|^{-n-it}$.

§§3-4 Generalized operators and maximal truncations

Standard kernels, Cauchy integrals, commutators, Calderón-Zygmund operators, Cotlar-type control, and almost-everywhere limits.

§§5-6 Vector-valued theory and further results

Banach-valued singular integrals, spaces of homogeneous type, strong singularities, and pseudo-differential operators.



**EXERCISE E01§1 · Proposition 5.2****Gradient condition implies the Hörmander condition**

CORE

SOURCE LOCUS: Chapter 5, Proposition 5.2, p. 92

Assume K is differentiable away from the origin and satisfies the gradient estimate (5.3) from the chapter.

- (a) For $|x| > 2|y|$, write $K(x - y) - K(x)$ using the fundamental theorem of calculus along the segment from x to $x - y$.
- (b) Use $|x - ty| \simeq |x|$ uniformly for $0 \leq t \leq 1$.
- (c) Obtain the pointwise bound $|K(x - y) - K(x)| \leq C|y|/|x|^{n+1}$.
- (d) Integrate over $|x| > 2|y|$ to prove the Hörmander condition (5.2).



Exercise E01

WORK PAGE 1

Gradient condition implies the Hörmander condition





Exercise E01

WORK PAGE 2

Gradient condition implies the Hörmander condition



**EXERCISE E02§1 · Theorem 5.1**

Rebuild the Calderón-Zygmund theorem from the Hilbert-transform proof

ADVANCED

SOURCE LOCUS: Chapter 5, Theorem 5.1, pp. 91–92

Let $Tf = K * f$ satisfy the L^2 multiplier bound (5.1) and the Hörmander condition (5.2).

- (a) Form the Calderón-Zygmund decomposition $f = g + b$ at height λ .
- (b) Estimate the good part using the L^2 bound.
- (c) For each bad cube Q_j with center c_j , use $\int b_j = 0$ to rewrite

$$Tb_j(x) = \int_{Q_j} [K(x - y) - K(x - c_j)]b_j(y) dy.$$

- (d) Integrate outside the enlarged cube and use (5.2) to finish the weak $(1, 1)$ estimate. Then explain interpolation and duality for all $1 < p < \infty$.



Exercise E02

WORK PAGE 1

Rebuild the Calderón-Zygmund theorem from the Hilbert-transform proof





Exercise E02

WORK PAGE 2

Rebuild the Calderón-Zygmund theorem from the Hilbert-transform proof





PRINCETON PROOF SHEET

Exercise E02

WORK PAGE 3

Rebuild the Calderón-Zygmund theorem from
the Hilbert-transform proof



**EXERCISE E03§1 · Proposition 5.3**

A Dini condition for homogeneous kernels

ADVANCED

SOURCE LOCUS: Chapter 5, Proposition 5.3, p. 93

Let $K(x) = \Omega(x')/|x|^n$ and let

$$\omega_\Omega(t) = \sup\{|\Omega(u_1) - \Omega(u_2)| : |u_1 - u_2| < t\}.$$

Assume $\int_0^1 \omega_\Omega(t) dt/t < \infty$.

- (a) Split $|K(x - y) - K(x)|$ into an angular difference and a radial-denominator difference.
- (b) Show the radial term is integrable on $|x| > 2|y|$.
- (c) Prove $|(x - y)' - x'| \leq C|y|/|x|$ there.
- (d) Convert the angular integral to the Dini integral and obtain the Hörmander condition.



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Exercise E03

WORK PAGE 1

A Dini condition for homogeneous kernels



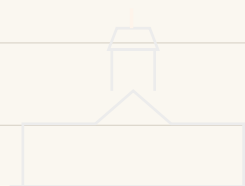


PRINCETON PROOF SHEET

Exercise E03

WORK PAGE 2

A Dini condition for homogeneous kernels





PRINCETON PROOF SHEET

Exercise E03

WORK PAGE 3

A Dini condition for homogeneous kernels



**EXERCISE E04§2 · conditions (5.7) and (5.9)**

The annular size condition and its weighted first moment

CORE

SOURCE LOCUS: Chapter 5, Proposition 5.5, p. 94

Assume

$$\int_{a < |x| < 2a} |K(x)| \, dx \leq B \quad (a > 0).$$

(a) Decompose $\{|x| < a\}$ into dyadic annuli.

(b) Prove

$$\int_{|x| < a} |x| |K(x)| \, dx \leq C B a.$$

(c) Conversely, show that such a first-moment estimate implies a uniform annular L^1 bound of the same type.

(d) Explain why this equivalence is useful in controlling the low-frequency part of a truncated Fourier transform.



Exercise E04

WORK PAGE 1

The annular size condition and its weighted first moment





Exercise E04

WORK PAGE 2

The annular size condition and its weighted first moment



**EXERCISE E05§2 · Proposition 5.5**

Uniform Fourier bounds for doubly truncated kernels

ADVANCED

SOURCE LOCUS: Chapter 5, Proposition 5.5, pp. 94–95

Let $K_{\varepsilon,R} = K \mathbf{1}_{\{\varepsilon < |x| < R\}}$ and assume (5.6)–(5.8).

- (a) Fix ξ and split $\widehat{K}_{\varepsilon,R}(\xi)$ at the scale $|\xi|^{-1}$.
- (b) Control the inner portion using cancellation (5.6) and the first-moment estimate equivalent to (5.7).
- (c) For the outer portion, choose z with $e^{2\pi i z \cdot \xi} = -1$ and average the original integral with its translate by z .
- (d) Bound the resulting difference by the Hörmander condition plus two annular error terms, obtaining a bound independent of ε , R and ξ .



Exercise E05

WORK PAGE 1

Uniform Fourier bounds for doubly truncated kernels





Exercise E05

WORK PAGE 2

Uniform Fourier bounds for doubly truncated kernels





Exercise E05

WORK PAGE 3

Uniform Fourier bounds for doubly truncated kernels



**EXERCISE E06§2 · Corollary 5.6**

Uniform L^p estimates for truncated convolutions

ADVANCED

SOURCE LOCUS: Chapter 5, Corollary 5.6, p. 95

Fill in the details the chapter leaves to the reader.

- (a) Use Proposition 5.5 and Plancherel to prove a uniform L^2 bound for $K_{\varepsilon,R} * f$.
- (b) Show that the truncated kernels satisfy a Hörmander condition with a constant independent of ε, R .
- (c) Apply Theorem 5.1 to deduce uniform strong (p, p) bounds and weak $(1, 1)$ bounds.
- (d) Identify carefully which constants depend only on the hypotheses of Proposition 5.5.



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Exercise E06

WORK PAGE 1

Uniform L^p estimates for truncated convolutions





PRINCETON PROOF SHEET

Exercise E06

WORK PAGE 2

Uniform L^p estimates for truncated convolutions





PRINCETON PROOF SHEET

Exercise E06

WORK PAGE 3

Uniform L^p estimates for truncated convolutions



**EXERCISE E07§2 · Proposition 5.7**

When does the principal-value distribution exist?

CORE

SOURCE LOCUS: Chapter 5, Proposition 5.7, pp. 96–97

Assume the annular bound (5.7).

- (a) Suppose p. v. K exists. Choose a Schwartz function equal to 1 on the unit ball and prove that $\int_{\varepsilon < |x| < 1} K(x) dx$ has a limit.
- (b) Conversely, assume that limit exists and denote it by L .
- (c) Derive the representation

$$\begin{aligned} \text{p. v. } K(\varphi) &= \varphi(0)L + \int_{|x| < 1} K(x)(\varphi(x) - \varphi(0)) dx \\ &\quad + \int_{|x| > 1} K(x)\varphi(x) dx. \end{aligned}$$

- (d) Prove both integrals converge and conclude that the principal-value distribution exists.



Exercise E07

WORK PAGE 1

When does the principal-value distribution exist?





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Exercise E07

WORK PAGE 2

When does the principal-value distribution exist?



**EXERCISE E08§2 · Example 5.9****The oscillatory kernel $|x|^{-n-it}$**

ADVANCED

SOURCE LOCUS: Chapter 5, Example 5.9, pp. 97–98

Fix $t \neq 0$ and consider $K(x) = |x|^{-n-it}$.

- (a) Verify the uniform annular L^1 bound and a gradient estimate of the type used in Proposition 5.5.
- (b) Compute the truncated radial integral over $\varepsilon < |x| < 1$ and show why it has no limit as $\varepsilon \rightarrow 0$.
- (c) Choose the subsequence $\varepsilon_k = e^{-2\pi k/t}$ (for the sign convention of the chapter) and verify that the oscillatory factor stabilizes.
- (d) Explain why the resulting “principal value” depends on the chosen renormalization and therefore is not the canonical homogeneous distribution.



Exercise E08

WORK PAGE 1

The oscillatory kernel $|x|^{-n-it}$





Exercise E08

WORK PAGE 2

The oscillatory kernel $|x|^{-n-it}$





Exercise E08

WORK PAGE 3

The oscillatory kernel $|x|^{-n-it}$



**EXERCISE E09§3 · Theorem 5.10**

Generalized Calderón-Zygmund operators

CORE

SOURCE LOCUS: Chapter 5, Theorem 5.10, p. 99

Let T be bounded on $L^2(\mathbb{R}^n)$ with off-support kernel representation $K(x, y)$ and assume the two integral Hörmander conditions (5.10)–(5.11).

- (a) Explain why only the off-support representation of T is needed in the Calderón-Zygmund decomposition proof.
- (b) Use (5.10) to obtain weak $(1, 1)$ boundedness of T .
- (c) Identify the kernel of the adjoint T^* and use (5.11) to obtain the corresponding weak estimate for T^* .
- (d) Finish the strong L^p range by interpolation and duality.



Exercise E09

WORK PAGE 1

Generalized Calderón-Zygmund operators





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Exercise E09

WORK PAGE 2

Generalized Calderón-Zygmund operators



**EXERCISE E10§3 · standard kernels**

Standard kernels satisfy the integral Hörmander conditions

CORE

SOURCE LOCUS: Chapter 5, conditions (5.12)–(5.14), p. 99

Suppose $K(x, y)$ satisfies the size estimate (5.12) and the Hölder regularity estimates (5.13)–(5.14) with exponent $\delta > 0$.

- (a) Fix y, z and integrate the difference estimate over $|x - y| > 2|y - z|$.
- (b) Use polar coordinates centered at y to show the integral is bounded independently of y, z .
- (c) Repeat the argument for the symmetric condition in the first variable.
- (d) Conclude that every standard kernel satisfies (5.10) and (5.11).



PRINCETON PROOF SHEET

Exercise E10

WORK PAGE 1

Standard kernels satisfy the integral
Hörmander conditions





Exercise E10

WORK PAGE 2

Standard kernels satisfy the integral
Hörmander conditions



**EXERCISE E11§3 · example (1)**

The Cauchy kernel on a Lipschitz graph is standard

ADVANCED

SOURCE LOCUS: Chapter 5, equations (5.15), pp. 99–100

Let $A : \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz and

$$K(x, y) = \frac{1}{x - y + i(A(x) - A(y))}.$$

- (a) Use the Lipschitz bound to prove $|K(x, y)| \leq C/|x - y|$.
- (b) Estimate $K(x, y) - K(x, z)$ when $|x - y| > 2|y - z|$.
- (c) Derive the analogous estimate in the first variable.
- (d) Conclude that K is a standard kernel with exponent $\delta = 1$, as stated in the chapter.

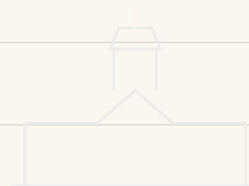


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Exercise E11

WORK PAGE 1

The Cauchy kernel on a Lipschitz graph is stan-
dard



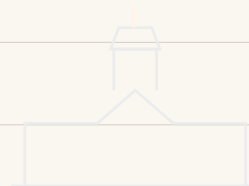


PRINCETON PROOF SHEET

Exercise E11

WORK PAGE 2

The Cauchy kernel on a Lipschitz graph is stan-
dard





Exercise E11

WORK PAGE 3

The Cauchy kernel on a Lipschitz graph is standard



**EXERCISE E12§3 · example (2)**

Calderón commutator kernels

ADVANCED

SOURCE LOCUS: Chapter 5, §3, p. 100

Let A be Lipschitz and define

$$K_k(x, y) = \left(\frac{A(x) - A(y)}{x - y} \right)^k \frac{1}{x - y}.$$

- (a) Prove the size bound $|K_k(x, y)| \leq C_k |x - y|^{-1}$.
- (b) Estimate the difference in the y variable using the Lipschitz property of A .
- (c) Obtain the corresponding estimate in the x variable.
- (d) Conclude that K_k is standard with $\delta = 1$; explain why the remaining issue for the operator is L^2 boundedness.



PRINCETON PROOF SHEET

Exercise E12

WORK PAGE 1

Calderón commutator kernels





Exercise E12

WORK PAGE 2

Calderón commutator kernels





Exercise E12

WORK PAGE 3

Calderón commutator kernels



**EXERCISE E13§4 · Proposition 5.13**

Operators with the same kernel differ by multiplication

ADVANCED

SOURCE LOCUS: Chapter 5, Proposition 5.13, pp. 101–102

The chapter leaves this proof to the reader. Let T_1, T_2 be Calderón-Zygmund operators with the same associated kernel and put $S = T_1 - T_2$.

- (a) Show that if f has compact support then $Sf(x) = 0$ for almost every $x \notin \text{supp } f$.
- (b) Use boundedness on L^2 and localization by characteristic functions of measurable sets to prove that S does not enlarge supports.
- (c) Deduce that S commutes with multiplication by characteristic functions.
- (d) Conclude that $Sf = a(x)f(x)$ for some $a \in L^\infty$ and identify the L^∞ bound from $\|S\|_{2 \rightarrow 2}$.



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Exercise E13

WORK PAGE 1

Operators with the same kernel differ by multiplication



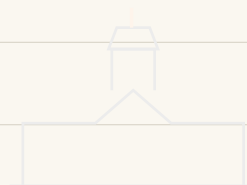


PRINCETON PROOF SHEET

Exercise E13

WORK PAGE 2

Operators with the same kernel differ by multiplication





PRINCETON PROOF SHEET

Exercise E13

WORK PAGE 3

Operators with the same kernel differ by multiplication



**EXERCISE E14§4 · Lemma 5.16****Kolmogorov's local L^u estimate**

CORE

SOURCE LOCUS: Chapter 5, Lemma 5.16, pp. 102–103

Let S be weak $(1, 1)$, let $0 < \nu < 1$, and let E have finite measure.

- (a) Use the distribution-function identity for $\int_E |Sf|^\nu$.
- (b) Bound the distribution function by the minimum of $|E|$ and $C\|f\|_1/\lambda$.
- (c) Split the λ integral at $\lambda_0 = C\|f\|_1/|E|$.
- (d) Compute both pieces and obtain the inequality stated in Lemma 5.16.



Exercise E14

WORK PAGE 1

Kolmogorov's local L^u estimate



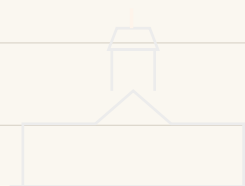


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Exercise E14

WORK PAGE 2

Kolmogorov's local L^u estimate



**EXERCISE E15§4 · Lemma 5.15**

Cotlar-type control of maximal truncations

ADVANCED

SOURCE LOCUS: Chapter 5, Lemma 5.15, pp. 102–104

Let T be a Calderón-Zygmund operator and $0 < \nu < 1$.

- (a) Fix a truncation scale ε and split $f = f\mathbf{1}_{2Q} + f\mathbf{1}_{(2Q)^c}$ for $Q = B(0, \varepsilon/2)$.
- (b) Use standard-kernel regularity to compare the far part of Tf at $z \in Q$ with the truncated value at the center.
- (c) Average the resulting pointwise inequality over Q .
- (d) Apply Lemma 5.16 to the local piece and derive

$$T^*f(x) \leq C_\nu \left(M(|Tf|^\nu)(x)^{1/\nu} + Mf(x) \right).$$



Exercise E15

WORK PAGE 1

Cotlar-type control of maximal truncations





Exercise E15

WORK PAGE 2

Cotlar-type control of maximal truncations





Exercise E15

WORK PAGE 3

Cotlar-type control of maximal truncations



**EXERCISE E16§4 · Theorem 5.14**

The maximal truncation theorem

ADVANCED

SOURCE LOCUS: Chapter 5, Theorem 5.14, pp. 102–105

Use Lemma 5.15 to prove Theorem 5.14.

- (a) For $1 < p < \infty$, first take $\nu = 1$ in the Cotlar estimate and deduce strong (p, p) boundedness of T^* .
- (b) For the weak endpoint, choose $0 < \nu < 1$ and split the level set of T^*f into a maximal-function term and a term involving $M(|Tf|^\nu)$.
- (c) Use the dyadic maximal operator as in the chapter and Lemma 5.16 to bound the second term.
- (d) Conclude that T^* is weak $(1, 1)$ and explain how this yields almost-everywhere convergence of truncations whenever the principal-value identity holds on a dense class.



PRINCETON PROOF SHEET

Exercise E16

WORK PAGE 1

The maximal truncation theorem





Exercise E16

WORK PAGE 2

The maximal truncation theorem





PRINCETON PROOF SHEET

Exercise E16

WORK PAGE 3

The maximal truncation theorem



**EXERCISE E17§5 · setup for Theorem 5.17**

Banach-valued L^p spaces and vector integration

CORE

SOURCE LOCUS: Chapter 5, §5, pp. 105–106

Let B be a separable Banach space.

- (a) State the chapter's notion of strong measurability and explain why $x \mapsto \|F(x)\|_B$ is measurable.
- (b) Define $L^p(B)$ and verify $\|f b\|_{L^p(B)} = \|f\|_p \|b\|_B$.
- (c) For finite sums $F = \sum_j f_j b_j$, define the Bochner integral as in the chapter and check compatibility with dual pairings against $b' \in B^*$.
- (d) Prove the Hölder estimate for the scalar pairing $\langle F, G \rangle$ with $G \in L^{p'}(B^*)$.



PRINCETON PROOF SHEET

Exercise E17

WORK PAGE 1

Banach-valued L^p spaces and vector integra-
tion





Exercise E17

WORK PAGE 2

Banach-valued L^p spaces and vector integration



**EXERCISE E18§5 · Theorem 5.17**

Vector-valued Calderón-Zygmund extension

ADVANCED

SOURCE LOCUS: Chapter 5, Theorem 5.17, pp. 106–107

Fill in the proof outline of Theorem 5.17 by following the scalar model.

- (a) Use condition (5.20) together with boundedness $L^r(A) \rightarrow L^r(B)$ to prove weak $(1, 1)$.
- (b) Interpolate to obtain boundedness for $1 < p < r$.
- (c) When A is reflexive, pass to the adjoint and use (5.21) to obtain the range $p > r$.
- (d) Explain the finite-dimensional subspace argument used in the chapter when $L^{p'}(A^*)$ is not known to equal $L^p(A)^*$.



PRINCETON PROOF SHEET

Exercise E18

WORK PAGE 1

Vector-valued Calderón-Zygmund extension





PRINCETON PROOF SHEET

Exercise E18

WORK PAGE 2

Vector-valued Calderón-Zygmund extension





Exercise E18

WORK PAGE 3

Vector-valued Calderón-Zygmund extension



**PROBLEM P01§6.2 · further results**

A more general rotational Dini condition

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.2, pp. 107–108

For $\rho \in O(n)$ define $\|\rho\| = \sup_{u \in S^{n-1}} |u - \rho u|$ and

$$\omega_1(t) = \sup_{\|\rho\| \leq t} \int_{S^{n-1}} |\Omega(\rho u) - \Omega(u)| d\sigma(u).$$

- (a) Compare this modulus with the pointwise modulus used in Proposition 5.3.
- (b) Explain why $\int_0^1 \omega_1(t) dt/t < \infty$ is weaker than pointwise Dini continuity.
- (c) Outline how rotational changes of variables enter the proof of the Hörmander condition.
- (d) Record the additional fact stated in the notes: this condition implies $\Omega \in L \log^+ L$.



Problem P01

WORK PAGE 1

A more general rotational Dini condition





Problem P01

WORK PAGE 2

A more general rotational Dini condition





Problem P01

WORK PAGE 3

A more general rotational Dini condition



**PROBLEM P02§6.3 · quasi-norm geometry**

Calderón-Zygmund theory for non-isotropic dilations

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.3, pp. 108–109

Let $\delta_t(x) = (t^{a_1}x_1, \dots, t^{a_n}x_n)$ with $a_j > 0$ and let $\|x\|$ be the associated quasi-norm from the chapter.

- (a) Verify $\|\delta_t x\| = t\|x\|$.
- (b) Explain why only a quasi-triangle inequality $\|x + y\| \leq L(\|x\| + \|y\|)$ is expected.
- (c) Formulate the analogues of the two integral Hörmander conditions using this quasi-distance.
- (d) Explain why the proof of Theorem 5.10 survives after replacing Euclidean balls and separation estimates by the quasi-metric geometry.



Problem P02

WORK PAGE 1

Calderón-Zygmund theory for nonisotropic dilations





Problem P02

WORK PAGE 2

Calderón-Zygmund theory for nonisotropic dilations





Problem P02

WORK PAGE 3

Calderón-Zygmund theory for nonisotropic dilations



**PROBLEM P03**§6.4 · Coifman-Weiss framework

Singular integrals on spaces of homogeneous type

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.4, pp. 109–110

Work in a space X with pseudo-metric ρ and doubling measure μ as described in the chapter notes.

- (a) State the pseudo-metric axioms and the doubling condition.
- (b) Explain how the doubling hypothesis replaces Euclidean volume growth in covering and Calderón-Zygmund decomposition arguments.
- (c) State the integral regularity condition (5.22) for an associated kernel.
- (d) Outline why an L^q bound for one $q > 1$ plus (5.22) yields weak $(1, 1)$ and strong L^p bounds for $1 < p < q$.



PRINCETON PROOF SHEET

Problem P03

WORK PAGE 1

Singular integrals on spaces of homogeneous type





Problem P03

WORK PAGE 2

Singular integrals on spaces of homogeneous type



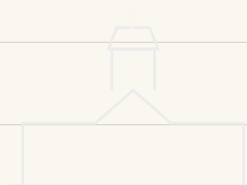


PRINCETON PROOF SHEET

Problem P03

WORK PAGE 3

Singular integrals on spaces of homogeneous type



**PROBLEM P04§6.7 · an operator-valued kernel**

Maximal functions as vector-valued singular integrals

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.7, p. 111

Let $\varphi \in \mathcal{S}$ be nonnegative and dominate the indicator of the unit ball. Set $B = \ell^\infty(\mathbb{Q}_+)$ and

$$K(x) = \{\varphi_t(x)\}_{t \in \mathbb{Q}_+}.$$

- (a) Show that convolution with this B -valued kernel has norm equal to the rational-scale maximal convolution operator $M_\varphi f$.
- (b) Verify boundedness $L^\infty(\mathbb{R}) \rightarrow L^\infty(B)$.
- (c) Use the Schwartz decay of φ to prove the operator-valued Hörmander condition.
- (d) Apply the vector-valued theorem to recover L^p boundedness of M_φ , and then compare M and M_φ as in the chapter.

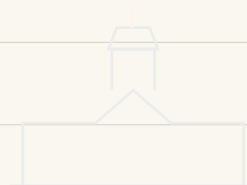


PRINCETON PROOF SHEET

Problem P04

WORK PAGE 1

Maximal functions as vector-valued singular
integrals





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Problem P04

WORK PAGE 2

Maximal functions as vector-valued singular
integrals



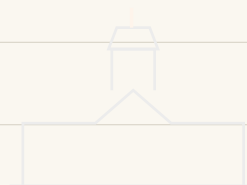


PRINCETON PROOF SHEET

Problem P04

WORK PAGE 3

Maximal functions as vector-valued singular
integrals



**PROBLEM P05**§6.8 · strongly singular integrals**Strongly singular kernels: what replaces Hörmander regularity?**

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.8, pp. 112–113

Consider the prototype multiplier/kernel pair described in §6.8 with oscillation $e^{i|\xi|^a}$ and power decay, $0 < a < 1$.

- (a) Record the two competing estimates emphasized in the notes: improved decay of the Fourier transform and stronger singular growth of the physical-space kernel.
- (b) Explain why the classical gradient/Hörmander condition fails for this prototype.
- (c) State qualitatively how Theorem 5.18 balances the exponents to determine the L^p range.
- (d) Explain the purpose of Theorem 5.19: stronger Fourier decay permits a weakened spatial smoothness hypothesis while retaining L^p and weak $(1, 1)$ control.



Problem P05

WORK PAGE 1

Strongly singular kernels: what replaces Hörmander regularity?





Problem P05

WORK PAGE 2

Strongly singular kernels: what replaces Hörmander regularity?





Problem P05

WORK PAGE 3

Strongly singular kernels: what replaces Hörmander regularity?



**PROBLEM P06§6.9 · symbols and kernels**

Pseudo-differential operators through the Calderón-Zygmund lens

CHALLENGE

SOURCE LOCUS: Chapter 5, §6.9, pp. 113–114

Consider the formal operator

$$Tf(x) = \int_{\mathbb{R}^n} \sigma(x, \xi) \widehat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi.$$

- (a) Identify the special cases in which σ depends only on ξ or only on x .
- (b) Rewrite T formally as an integral operator with kernel $K(x, x - y)$, following equation (5.24).
- (c) State the symbol class S^m estimate given in the chapter and explain why S^0 symbols lead to kernels satisfying Hörmander-type regularity plus L^2 boundedness.
- (d) Summarize the symbolic-calculus idea for compositions and explain how ellipticity leads to a parametrix: an inverse modulo a lower-order error.



PRINCETON PROOF SHEET

Problem P06

WORK PAGE 1

Pseudo-differential operators through the
Calderón-Zygmund lens





PRINCETON PROOF SHEET

Problem P06

WORK PAGE 2

Pseudo-differential operators through the
Calderón-Zygmund lens



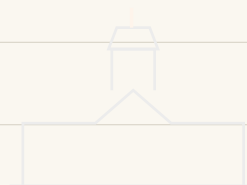


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Problem P06

WORK PAGE 3

Pseudo-differential operators through the
Calderón-Zygmund lens





CHAPTER VI

H^1 and BMO

ATOMS • OSCILLATION • DUALITY

EXERCISE E01 §1 • Definition 6.1**Atomic normalization and cancellation**

SOURCE LOCUS: CHAPTER 6, DEFINITION 6.1, p. 115

Let $Q \subset \mathbb{R}^n$ be a cube and let a be supported in Q , satisfy

$$\int_Q a(x) \, dx = 0, \quad \|a\|_\infty \leq |Q|^{-1}.$$

- a. Prove that $\|a\|_1 \leq 1$ and $\|a\|_2 \leq |Q|^{-1/2}$.
- b. If a is rescaled by $a_r(x) = r^{-n}a(x/r)$, identify the support, cancellation, and L^∞ normalization.
- c. Explain why the cancellation condition is the mechanism that makes the far-field singular integral estimate converge.

CORE



PRINCETON PROOF SHEET

Exercise E01

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E01

WORK PAGE 2



EXERCISE E02 §1 • Proposition 6.2**A singular integral maps an atom into L^1**

SOURCE LOCUS: CHAPTER 6, PROPOSITION 6.2, PP. 115–116

Assume T is bounded on $L^2(\mathbb{R}^n)$ and its kernel satisfies the integral smoothness condition of Chapter 5. Let a be an atom supported on Q with center c_Q .

a. Choose a fixed enlargement Q^* of Q and estimate $\int_{Q^*} |Ta|$ by the L^2 bound.

b. For $x \notin Q^*$ use $\int_Q a = 0$ to write

$$Ta(x) = \int_Q (K(x, y) - K(x, c_Q))a(y) dy.$$

c. Integrate the smoothness estimate in x and conclude $\|Ta\|_1 \leq C$ uniformly in the atom.

d. Extend the estimate from atoms to finite atomic sums and identify the resulting $H_{\text{at}}^1 \rightarrow L^1$ operator bound.

CORE



PRINCETON PROOF SHEET

Exercise E02

WORK PAGE 1





PRINCETON PROOF SHEET

Exercise E02

WORK PAGE 2



EXERCISE E03 §2 • Proposition 6.5**Three useful formulations of BMO**

SOURCE LOCUS: CHAPTER 6, PROPOSITION 6.5, p. 117

For locally integrable f , let

$$\|f\|_* := \sup_Q \frac{1}{|Q|} \int_Q |f - f_Q|.$$

Prove the equivalence, up to an absolute dimensional constant, between $\|f\|_*$ and

$$\sup_Q \inf_{a \in \mathbb{C}} \frac{1}{|Q|} \int_Q |f(x) - a| dx.$$

Then prove the pointwise estimate

$$M^\#(|f|)(x) \leq 2M^\#f(x).$$

Finally, explain carefully why $f \in L^\infty$ implies $f \in \text{BMO}$, while BMO does not impose boundedness. CORE



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EXERCISE E04 §2 • Theorem 6.6**Bounded compact support functions go into BMO**

SOURCE LOCUS: CHAPTER 6, THEOREM 6.6, PP. 118–119

Let T satisfy the hypotheses of the theorem and let $f \in L^\infty$ have compact support. For a cube Q with center c_Q , split $f = h + \ell$ with $h = f\chi_{Q^*}$ and choose $a = T\ell(c_Q)$.

- a. Show that the local contribution is controlled by the L^2 boundedness of T .
- b. Use the kernel regularity in the first variable to estimate

$$\frac{1}{|Q|} \int_Q |T\ell(x) - T\ell(c_Q)| \, dx.$$

- c. Combine the two pieces and prove $\|Tf\|_* \leq C\|f\|_\infty$.

CORE



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EXERCISE E05 §2 • Example 6.7**The Hilbert transform of sgn**

SOURCE LOCUS: CHAPTER 6, EXAMPLE 6.7, PP. 120

Let $f(x) = \operatorname{sgn}(x)$ and use the Hilbert-transform normalization in the source. Fix $a > 0$ and work first for $|x| < a/2$.

- a. Write the principal-value integral for $Hf(x)$ by splitting the integrals across the singularity.
- b. Compute the truncated pieces explicitly and pass to the limit.
- c. Show that, modulo an additive constant,

$$H(\operatorname{sgn})(x) = -\frac{2}{\pi} \log |x|.$$

- d. Explain why this gives a concrete unbounded BMO function.

ADVANCED



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EXERCISE E06 §4 • Theorem 6.11**John–Nirenberg by iterative cube decompositions**

SOURCE LOCUS: CHAPTER 6, THEOREM 6.11, PP. 123–124

Assume $\|f\|_* = 1$. Let Q be a cube and perform the dyadic Calderon–Zygmund decomposition of $f - f_Q$ on Q at height 2.

- Prove the first generation cubes have total measure at most a fixed fraction of $|Q|$.
- Show that outside the second-generation cubes, $|f - f_Q|$ is bounded by a dimensional multiple of 2^2 .
- Iterate the construction to obtain generations $\{Q_{N,j}\}$ with total measure decaying geometrically in N .
- For arbitrary $A > 0$, choose N with $N2^{n+1} < A < (N + 1)2^{n+1}$ and deduce

$$|\{x \in Q : |f - f_Q| > A\}| \leq C_1 e^{-C_2 A} |Q|.$$

ADVANCED



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EXERCISE E07 §3 • Theorem 6.8**Interpolation with an $L^\infty \rightarrow \text{BMO}$ endpoint**

SOURCE LOCUS: CHAPTER 6, THEOREM 6.8, PP. 121–122

Let T be linear, bounded on L^{p_0} for some $1 < p_0 < \infty$, and bounded from L^∞ into BMO.

- a. For compactly supported f , show $M^\#(Tf)$ is bounded on L^{p_0} .
- b. Explain how the good- λ inequality yields a norm estimate for the maximal function in terms of the sharp maximal function.
- c. Apply Marcinkiewicz interpolation to $M^\# \circ T$ and deduce L^p boundedness of T for all $p_0 < p < \infty$.

ADVANCED



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PROBLEM P01 §5 • Theorem 6.15**Why H^1 and BMO are dual spaces**

SOURCE LOCUS: CHAPTER 6, THEOREM 6.15, p. 129

Treat the statement as a proof project. Assume the atomic description of H^1 and the atomic estimate for singular integrals are available.

- a. For $b \in \text{BMO}$ and an atom a supported on Q , show that $|\int ab|$ is controlled by $\|b\|_*$.
- b. Extend the estimate from atoms to finite atomic sums and obtain a bounded functional on H^1 .
- c. Identify the obstruction to using the naive pointwise pairing for arbitrary H^1 functions and explain where a limiting argument is needed.
- d. State precisely the converse representation of every bounded functional on H^1 by a BMO function, including the quotient by constants.

CHALLENGE



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PROBLEM P02 §5.6 • Theorem 6.17**The endpoint** $I_\alpha : L^{n/\alpha} \rightarrow \text{BMO}$

SOURCE LOCUS: CHAPTER 6, THEOREMS 6.17–6.18, p. 130

Let $0 < \alpha < n$ and I_α be the fractional integral. Use the source theorem

$$C_1 M_\alpha f(x) \leq M^\#(I_\alpha f)(x) \leq C_2 M_\alpha f(x)$$

to build the endpoint argument.

- a. Explain the scaling that singles out $p = n/\alpha$.
- b. Show that $f \in L^{n/\alpha}$ gives $M_\alpha f \in L^\infty$.
- c. Deduce $I_\alpha f \in \text{BMO}$ and record the dependence of the BMO seminorm on $\|f\|_{n/\alpha}$.
- d. Compare this endpoint with the false expectation $I_\alpha : L^{n/\alpha} \rightarrow L^\infty$.

CHALLENGE



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PROBLEM P03 §5.8 • Theorem 6.20**BMO through commutators**

SOURCE LOCUS: CHAPTER 6, THEOREM 6.20, PP. 131–132

Let T be a Calderon–Zygmund singular integral with standard kernel and define

$$[b, T]f = bTf - T(bf).$$

a. Derive the off-support kernel formula

$$[b, T]f(x) = \int (b(x) - b(y))K(x - y)f(y) dy.$$

b. Explain why $b \in L^\infty$ is sufficient for L^p boundedness.

c. Explain why the BMO condition is weaker and nevertheless sufficient.

d. State the converse and indicate why cancellation between the two terms is the key obstruction to testing boundedness by b alone.

CHALLENGE



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CHAPTER VII

Weighted Inequalities

A_p • REVERSE HOLDER • EXTRAPOLATION

EXERCISE E08 §1 • Theorem 7.1**From weighted weak type to the A_p condition**

SOURCE LOCUS: CHAPTER 7, THEOREM 7.1, PP. 133–136

Assume the Hardy–Littlewood maximal operator satisfies

$$w(\{Mf > \lambda\}) \leq \frac{C}{\lambda^p} \int_{\mathbb{R}^n} |f|^p w.$$

- a. Test the inequality on $f = \chi_Q$ and obtain a two-set estimate for $S \subset Q$.
- b. Treat separately $p = 1$ and $1 < p < \infty$.
- c. In the case $p = 1$, derive the A_1 condition and the equivalent pointwise inequality $Mw \leq Cw$ a.e.
- d. In the case $p > 1$, choose the source test function involving $w^{-p'}\chi_Q$ and derive the A_p condition.

CORE



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Exercise E08

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EXERCISE E09 §1 • Proposition 7.2**Structural identities inside the A_p scale**

SOURCE LOCUS: CHAPTER 7, PROPOSITION 7.2, PP. 136–137

Let $1 < p < q < \infty$.a. Prove $A_p \subset A_q$.

b. Prove the duality relation

$$w \in A_p \iff w^{1-p'} \in A_{p'}.$$

c. Show that if $w_0, w_1 \in A_1$, then $w = w_0 w_1^{1-p} \in A_p$.d. Verify the model power weight $w(x) = |x|^\alpha$ lies in $A_p(\mathbb{R}^n)$ exactly for

$$-n < \alpha < n(p-1).$$

CORE



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Exercise E09

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EXERCISE E10 §2 • Theorems 7.3–7.5**Reverse Holder self-improvement**

SOURCE LOCUS: CHAPTER 7, THEOREMS 7.3–7.5, PP. 137–139

Assume $w \in A_p$.

- a. State the reverse Holder estimate in a form that quantitatively improves the integrability of w .
- b. Prove the measure-decay lemma: if $S \subset Q$ and $|S| \leq \alpha|Q|$, then $w(S) \leq \beta w(Q)$ for suitable $\beta < 1$ depending on the A_p data and α .
- c. Iterate the decay over Calderon–Zygmund generations and recover a reverse Holder exponent $1 + \varepsilon$.
- d. Explain how the reverse Holder estimate implies the existence of $q < p$ with $w \in A_q$, thereby closing the proof of the strong maximal-function theorem.

ADVANCED



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Exercise E10

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EXERCISE E11 §3 • **Theorem 7.7****Constructing A_1 weights from the maximal function**

SOURCE LOCUS: CHAPTER 7, THEOREM 7.7, p. 140

Let $0 < \delta < 1$ and suppose $Mf < \infty$ a.e. Prove that

$$w = (Mf)^\delta$$

is an A_1 weight with a constant depending only on δ and n .

- a. Reduce the claim to a local estimate on a fixed cube Q .
- b. Decompose $f = f\chi_{2Q} + f\chi_{\mathbb{R}^n \setminus 2Q}$.
- c. Control the far part uniformly on Q using the geometry of cubes.
- d. Use weak $(1, 1)$ and Kolmogorov's inequality on the local part to show

$$\frac{1}{|Q|} \int_Q (Mf)^\delta \leq C(Mf(x))^\delta$$

for a.e. $x \in Q$.

ADVANCED



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Exercise E11

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EXERCISE E12 §3 • Theorem 7.8**Rubio de Francia extrapolation as a proof machine**

SOURCE LOCUS: CHAPTER 7, THEOREM 7.8, PP. 140–142

Let T be a sublinear operator and suppose that for some $r \in (1, \infty)$, T is bounded on $L^r(w)$ for every $w \in A_r$, with the quantitative constant depending on the A_r characteristic.

- a. Describe the role of the A_1 majorant constructed by the iteration behind Rubio de Francia's algorithm.
- b. For a target exponent p , explain how the algorithm converts an arbitrary f into a controlled majorant whose weighted norm can be tested at exponent r .
- c. Use duality and the factorization properties of A_p weights to pass from the fixed exponent r to arbitrary $1 < p < \infty$.
- d. State the final extrapolation theorem precisely.

CHALLENGE



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Exercise E12

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Exercise E12

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EXERCISE E13 §4 • Theorem 7.11**Weighted Calderon–Zygmund boundedness**

SOURCE LOCUS: CHAPTER 7, THEOREMS 7.11–7.13, PP. 146–148

Let T be a Calderon–Zygmund singular integral and let $w \in A_p$, $1 < p < \infty$.

- a. Combine the unweighted Calderon–Zygmund theory with weighted maximal estimates to explain the architecture of the proof.
- b. Track where the A_p self-improvement enters.
- c. Derive the strong estimate $\|Tf\|_{L^p(w)} \leq C_{p,[w]_{A_p}} \|f\|_{L^p(w)}$.
- d. For $w \in A_1$, explain the corresponding weighted weak $(1, 1)$ statement for T .

ADVANCED



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EXERCISE E14 §4 • Corollary 7.13**Maximal truncations under A_p weights**

SOURCE LOCUS: CHAPTER 7, COROLLARY 7.13, p. 148

Let

$$T^*f(x) = \sup_{\varepsilon > 0} |T_\varepsilon f(x)|$$

be the maximal truncation of a Calderon–Zygmund operator.

a. Start from Cotlar’s inequality

$$T^*f \leq C_\nu \left(M(|Tf|^\nu)^{1/\nu} + Mf \right), \quad 0 < \nu < 1.$$

b. For $1 < p < \infty$, combine the weighted maximal theorem and the weighted singular-integral estimate.

c. For $w \in A_1$, identify the additional weak-type inputs needed to obtain weighted weak $(1, 1)$.

d. State the almost-everywhere convergence consequence for truncated singular integrals.

CORE



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Exercise E14

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PROBLEM P04 §5 • two-weight frontier**Weak type versus strong type for two weights**

SOURCE LOCUS: CHAPTER 7, NOTES AND FURTHER RESULTS, §5.9, pp. 154–155

Let $1 < p < \infty$ and let (u, v) be a pair of weights.

- a. Write the two-weight A_p condition and explain how it reduces to the usual A_p condition when $u = v$.
- b. Explain why the weak-type maximal inequality is characterized by this condition.
- c. Explain, at the level of proof strategy, why the same condition is not sufficient for the strong (p, p) inequality.
- d. State the testing characterization for the strong maximal-operator inequality in the form given in the source notes.

CHALLENGE



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Problem P04

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FINAL SYNTHESIS

From cancellation to oscillation, from oscillation to weight

Organize a proof map linking the following chain:

atom cancellation $\longrightarrow H^1 \rightarrow L^1, \quad L^\infty \rightarrow \text{BMO},$

John–Nirenberg $\longrightarrow A_p$ self-improvement,

weighted singular-integral bounds.

For each arrow, write the exact estimate you would need and identify the theorem in Chapters VI–VII that supplies it.



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Proof work

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Proof work

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Proof work

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END OF CHAPTERS VI–VII

Work slowly. Preserve the hypotheses. Respect the cancellations.



Chapter Map

LARGE-TYPE WORKING ATLAS

§1 Some vector-valued inequalities

Extension of scalar Calderón–Zygmund bounds, interval projections, and the weighted vector-valued mechanism.

§2 Littlewood–Paley theory

Dyadic blocks, smooth cutoffs, square functions, product decompositions, and norm equivalence.

§3–4 Multipliers

Hörmander regularity, Sobolev control, Marcinkiewicz variation, and higher-dimensional dyadic arguments.

§5 Bochner–Riesz multipliers

Critical index, positive range, kernel asymptotics, and the role of restriction phenomena.

§6–7 Singular integrals and curved averages

Dyadic decomposition, rough homogeneous kernels, square functions, and the Hilbert transform/maximal function along a parabola.



§8 Notes and further results

Arbitrary interval inequalities, the ball multiplier, restriction theory, weighted multipliers, and spherical maximal operators.



CHAPTER VIII

Littlewood–Paley Theory and Multipliers

Dyadic scale, square functions, multiplier structure, and curved singular operators

The chapter begins from vector-valued singular-integral estimates and builds toward Littlewood–Paley decompositions, Hörmander and Marcinkiewicz multiplier theorems, Bochner–Riesz theory, and the Hilbert transform and maximal function along a parabola.

SOURCE RANGE: Chapter 8, pp. 157–194.





EXERCISE E01 CORE

Vector-valued Calderón–Zygmund extension

SOURCE LOCUS: Theorem 8.1, Chapter 8, §1, pp. 157–158

CORE

Let T be a convolution operator which is bounded on $L^2(\mathbb{R}^n)$ and whose kernel satisfies the Hörmander condition. Reconstruct the vector-valued argument for sequences $\{f_j\}$.

- (a) Explain why the sequence operator $\{f_j\} \mapsto \{Tf_j\}$ has operator-valued kernel $K(x)I$.
- (b) Verify the vector-valued Hörmander condition by reducing the operator norm to the scalar kernel difference.
- (c) State the resulting $L^p(\ell^r)$ estimate for $1 < p, r < \infty$ and the corresponding $p = 1$ endpoint in the notation of the chapter.



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Exercise E01

Vector-valued Calder' on-Zygmund extension

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Exercise E01

Vector-valued Calder´on-Zygmund extension

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EXERCISE E02 CORE

Interval projections from the Hilbert transform

SOURCE LOCUS: Corollary 8.2, Chapter 8, §1, p. 158

CORE

For a sequence of finite or infinite intervals $I_j \subset \mathbb{R}$, define

$$\widehat{S_j f}(\xi) = \mathbf{1}_{I_j}(\xi) \widehat{f}(\xi).$$

For $I_j = (a_j, b_j)$, use the Hilbert-transform identity indicated in the chapter to express $S_j f$ through translated Hilbert transforms.

- (a) Derive the representation of $S_j f$ from the multiplier $\mathbf{1}_{(a_j, b_j)}$.
- (b) Deduce the vector-valued estimate for $\{S_j f_j\}$.
- (c) Identify exactly where Theorem 8.1 enters.



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Exercise E02

Interval projections from the Hilbert transform

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Exercise E02

Interval projections from the Hilbert transform

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EXERCISE E03 CORE

The Littlewood–Paley square-function equivalence

CORE

SOURCE LOCUS: Theorem 8.4, Chapter 8, §2, pp. 159–161

Let

$$I_j = (-2^{j+1}, -2^j] \cup [2^j, 2^{j+1}), \quad \widehat{S_j f} = \mathbf{1}_{I_j} \widehat{f}.$$

Prove the two-sided estimate

$$c_p \|f\|_{L^p} \leq \left\| \left(\sum_{j \in \mathbb{Z}} |S_j f|^2 \right)^{1/2} \right\|_{L^p} \leq C_p \|f\|_{L^p}, \quad 1 < p < \infty.$$

Your solution should reproduce the chapter's logical chain through the smooth-cutoff theorem and the polarization argument.



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Exercise E03

The Littlewood–Paley square-function equivalence

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Exercise E03

The Littlewood–Paley square-function equivalence

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EXERCISE E04 ADVANCED

Smooth dyadic cutoffs and the partition of frequency space

ADVANCED

SOURCE LOCUS: Theorem 8.5–8.6, Chapter 8, §2, pp. 160–162

Let $\psi \in \mathcal{S}(\mathbb{R}^n)$ satisfy $\psi(0) = 0$, and set $\psi_j(\xi) = \psi(2^{-j}\xi)$. Define S_j by $\widehat{S_j f} = \psi_j \widehat{f}$.

- (a) Explain why $S_j f = W_j * f$ for the inverse Fourier transform W_j .
- (b) Show that the kernel sequence satisfies a vector-valued Hörmander condition by estimating $\|W_j(x)\|_{\ell^2}$.
- (c) Prove the upper square-function bound.
- (d) Assume $\sum_j |\psi(2^{-j}\xi)|^2 = C$ for $\xi \neq 0$. Explain how the reverse inequality is recovered.



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Exercise E04

Smooth dyadic cutoffs and the partition of frequency space

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Exercise E04

Smooth dyadic cutoffs and the partition of frequency space

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Exercise E04

Smooth dyadic cutoffs and the partition of frequency space

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**EXERCISE E05 ADVANCED****A dyadic square function on the plane**

SOURCE LOCUS: Theorem 8.7, Chapter 8, §2, pp. 162–163

ADVANCED

For $f \in L^p(\mathbb{R}^2)$, define the product dyadic pieces by applying the one-dimensional decomposition in each coordinate. Reconstruct the proof of the estimate

$$\left\| \left(\sum_{j,k} |S_j^{(1)} S_k^{(2)} f|^2 \right)^{1/2} \right\|_p \asymp_p \|f\|_p, \quad 1 < p < \infty.$$

Emphasize where Fubini's theorem is used and why the one-dimensional argument can be iterated.



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Exercise E05

A dyadic square function on the plane

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Exercise E05

A dyadic square function on the plane

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Exercise E05

A dyadic square function on the plane

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EXERCISE E06 ADVANCED

Hörmander's multiplier theorem

SOURCE LOCUS: Theorem 8.10 and Corollary 8.11, Chapter 8, §3, pp. 164–166

ADVANCED

Let $\psi \in C_c^\infty$ be radial, supported in $\{1/2 < |\xi| < 2\}$, with

$$\sum_{j \in \mathbb{Z}} |\psi(2^{-j}\xi)|^2 = 1, \quad \xi \neq 0.$$

Suppose that for some $a > n/2$,

$$\sup_{j \in \mathbb{Z}} \|m(2^j \cdot) \psi\|_{L_a^2} < \infty.$$

- (a) Explain the role of the smooth Littlewood–Paley decomposition.
- (b) Isolate the weighted L^2 estimate for the localized multipliers.
- (c) Combine the vector-valued estimate with Littlewood–Paley theory to conclude L^p boundedness for $1 < p < \infty$.
- (d) Recast the hypothesis in the derivative form stated in Corollary 8.11.



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Exercise E06

Hörmander’s multiplier theorem

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Exercise E06

Hörmander’s multiplier theorem

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Exercise E06

Hörmander’s multiplier theorem

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EXERCISE E07 ADVANCED

Example 8.12: model multipliers

SOURCE LOCUS: Example 8.12, Chapter 8, §3, p. 166

ADVANCED

Work through both examples explicitly.

- (a) For $m(\xi) = |\xi|^{it}$, $t \in \mathbb{R}$, verify the derivative estimates required by Corollary 8.11.
- (b) Suppose m is homogeneous of degree 0 and of class C^k on the unit sphere with $k > [n/2]$. Explain why the homogeneity converts the sphere regularity into the required global derivative estimates away from the origin.
- (c) Compare the scale-invariance here with the dyadic localization used in Theorem 8.10.



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Exercise E07

Example 8.12: model multipliers

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Exercise E07

Example 8.12: model multipliers

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Exercise E07

Example 8.12: model multipliers

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EXERCISE E08 CORE

Marcinkiewicz variation on dyadic intervals

SOURCE LOCUS: Theorem 8.13, Chapter 8, §4, pp. 166–167

CORE

Let m be bounded on $\mathbb{R}$ and assume its total variation is uniformly bounded on each dyadic interval. Reconstruct the proof that m is an L^p multiplier, $1 < p < \infty$.

- (a) Decompose m into dyadic interval pieces.
- (b) Write the piecewise multiplier using the interval-endpoint values and the Stieltjes measure dm .
- (c) Apply the Hilbert-transform interval estimates to the truncated pieces.
- (d) Use Minkowski and the vector-valued Littlewood–Paley theorem to sum the dyadic contributions.



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Exercise E08

Marcinkiewicz variation on dyadic intervals

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Exercise E08

Marcinkiewicz variation on dyadic intervals

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**PROBLEM P01 ADVANCED****The Bochner–Riesz positive range**

SOURCE LOCUS: Theorem 8.15 and Lemma 8.16, Chapter 8, §5, pp. 168–172

ADVANCED

For $a > 0$, consider the Bochner–Riesz multiplier

$$T_R^a \widehat{f}(\xi) = \left(1 - \frac{|\xi|^2}{R^2}\right)_+^a \widehat{f}(\xi).$$

Reconstruct the positive part of the chapter’s argument.

- (a) Decompose the singularity near $|\xi| = 1$ into dyadic annuli.
- (b) Explain the normalization leading to the operators T_k .
- (c) Use the shell estimate from Lemma 8.16 and sum the resulting geometric series.
- (d) State the critical index $(n - 1)/2$ and the L^p range encoded by

$$\left| \frac{1}{p} - \frac{1}{2} \right| < \frac{2a + 1}{2n}.$$



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Problem P01

The Bochner-Riesz positive range

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Problem P01

The Bochner-Riesz positive range

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Problem P01

The Bochner-Riesz positive range

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PROBLEM P02 CHALLENGE

Bessel asymptotics and the negative Bochner–Riesz range

CHALLENGE

SOURCE LOCUS: Lemmas 8.17–8.18 and the proof of Theorem 8.15, Chapter 8, §5, pp. 171–172

Let $m_a(\xi) = (1 - |\xi|^2)_+^a$ and let $K_a = \widehat{m_a}$. The chapter gives a Bessel-function representation of K_a and uses its asymptotics to force a restriction on p .

- (a) Explain why a compactly supported multiplier on L^p must itself lie in L^p .
- (b) Starting from the Bessel representation in Lemma 8.18, isolate the behavior of $K_a(x)$ as $|x| \rightarrow 0$ and as $|x| \rightarrow \infty$.
- (c) Determine the resulting integrability restriction on p .
- (d) Explain how duality gives the complementary forbidden range.



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Problem P02

Bessel asymptotics and the negative Bochner-Riesz range

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Problem P02

Bessel asymptotics and the negative Bochner-Riesz range

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Problem P02

Bessel asymptotics and the negative Bochner-Riesz range

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Problem P02

Bessel asymptotics and the negative Bochner-Riesz range

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EXERCISE E09 ADVANCED

Dyadic singular-integral decomposition

SOURCE LOCUS: Theorem 8.19, Chapter 8, §6, pp. 172–176

ADVANCED

Let $K \in L^1(\mathbb{R}^n)$ be compactly supported with mean zero and assume the decay condition from Theorem 8.19. Define $K_j(x) = 2^{-jn}K(2^{-j}x)$ and

$$Tf = \sum_{j \in \mathbb{Z}} K_j * f.$$

- (a) Prove the L^2 estimate by Plancherel and the cancellation of K .
- (b) Introduce the annular frequency decomposition and define the pieces T_k .
- (c) Explain the L^2 decay $\|T_k f\|_2 \lesssim 2^{-\alpha|k|} \|f\|_2$.
- (d) Reconstruct the weak $(1, 1)$ estimate needed for interpolation.
- (e) Sum in k to obtain the L^p result.



PRINCETON PROOF SHEET

Exercise E09

Dyadic singular-integral decomposition

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Exercise E09

Dyadic singular-integral decomposition

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Exercise E09

Dyadic singular-integral decomposition

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EXERCISE E10 ADVANCED

Rough homogeneous singular integrals

SOURCE LOCUS: Corollary 8.21, Chapter 8, §6, pp. 175–177

ADVANCED

Suppose $\Omega \in L^q(S^{n-1})$, $q > 1$, and $\int_{S^{n-1}} \Omega d\sigma = 0$. Consider

$$Tf(x) = \text{p. v.} \int_{\mathbb{R}^n} \frac{\Omega(y')}{|y|^n} f(x - y) dy.$$

- (a) Reconstruct the oscillatory integral estimate used in Lemma 8.20.
- (b) Verify the exponent restriction coming from the integral in the sphere variable.
- (c) Explain how the dyadic decomposition converts the rough singular integral into the framework of Theorem 8.19.
- (d) State the final L^p boundedness range.



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Exercise E10

Rough homogeneous singular integrals

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Exercise E10

Rough homogeneous singular integrals

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Exercise E10

Rough homogeneous singular integrals

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EXERCISE E11 CORE

Van der Corput's lemma

SOURCE LOCUS: Lemma 8.27, Chapter 8, §7, pp. 183–184

CORE

Let $I(a, b) = \int_a^b e^{ih(t)} dt$.

- (a) Prove the first estimate assuming $|h'(t)| \geq \lambda > 0$ and h' monotone.
- (b) Prove the second estimate when $h \in C^k$ and $|h^{(k)}(t)| \geq \lambda > 0$.
- (c) For $k = 2$, carry out the interval decomposition around the possible zero of h' and optimize the splitting parameter.



PRINCETON PROOF SHEET

Exercise E11

Van der Corput’s lemma

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Exercise E11

Van der Corput’s lemma

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PROBLEM P03 CHALLENGE

Hilbert transform and maximal function along a parabola

CHALLENGE

SOURCE LOCUS: Lemma 8.28 and Corollary 8.29, Chapter 8, §7, pp. 183–184

Define

$$H_r f(x_1, x_2) = \text{p. v.} \int_{\mathbb{R}} f(x_1 - t, x_2 - t^2) \frac{dt}{t},$$

$$M_r f(x_1, x_2) = \sup_{h>0} \frac{1}{2h} \int_{-h}^h |f(x_1 - t, x_2 - t^2)| dt.$$

Reconstruct the proof that both operators are bounded on $L^p(\mathbb{R}^2)$, $1 < p < \infty$.

- (a) Identify the dyadic measures σ_j and μ_j .
- (b) Prove the oscillatory estimate of Lemma 8.28.
- (c) Check the hypotheses of Theorem 8.26 and explain the one-dimensional maximal-function reduction.
- (d) Explain how Theorem 8.25 finishes the argument for H_r .



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Problem P03

Hilbert transform and maximal function along a parabola

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Problem P03

Hilbert transform and maximal function along a parabola

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Problem P03

Hilbert transform and maximal function along a parabola

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Problem P03

Hilbert transform and maximal function along a parabola

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**PROBLEM P04 CHALLENGE****Arbitrary intervals and the ball multiplier****CHALLENGE**

SOURCE LOCUS: Theorems 8.30 and 8.32, Chapter 8, §8, pp. 186–188

- (a) State the arbitrary-interval Littlewood–Paley inequality for $p > 2$ and explain why the chapter warns that the analogue fails for $1 < p < 2$.
- (b) For the characteristic function of a Euclidean ball in $\mathbb{R}^n$, $n > 2$, explain the theorem that rules out multiplier boundedness on every L^p , $p \neq 2$.
- (c) Relate these two statements to the distinction between frequency decompositions with disjoint intervals and the curved boundary of a ball.



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Problem P04

Arbitrary intervals and the ball multiplier

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Problem P04

Arbitrary intervals and the ball multiplier

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Problem P04

Arbitrary intervals and the ball multiplier

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CHAPTER IX

The $T1$ Theorem

Cotlar, Carleson measures, weak boundedness, and the David–Journé architecture

The chapter returns to the L^2 boundedness problem for Calderón–Zygmund operators and organizes the answer around Cotlar’s lemma, Carleson measures, the weak boundedness property, the $T1$ theorem, and a paraproduct construction.

SOURCE RANGE: Chapter 9, pp. 195–212.





EXERCISE E12 CORE

Cotlar's lemma

SOURCE LOCUS: Lemma 9.1, Chapter 9, §1, pp. 195–197

CORE

Let H be a Hilbert space and $\{T_j\}$ a sequence of bounded operators. Reconstruct the proof of Cotlar's lemma from the SS^* argument.

- (a) Define a finite partial sum $S = \sum_{j=n}^m T_j$ and express $\|S\|$ in terms of powers of SS^* .
- (b) Explain the two ways of estimating each operator product in the expansion of $(SS^*)^k$.
- (c) Use the geometric-mean step and successive summation to obtain the final bound in terms of $\sum_i a(i)^{1/2}$.
- (d) Explain why the same hypotheses imply convergence of the series on all of H .



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Exercise E12

Cotlar’s lemma

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Exercise E12

Cotlar’s lemma

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Exercise E12

Cotlar’s lemma

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EXERCISE E13 CORE

Example 9.2: dyadic Hilbert-transform truncations

CORE

SOURCE LOCUS: Example 9.2, Chapter 9, §1, pp. 196–197

For

$$T_j f(x) = \int_{2^j < |t| \leq 2^{j+1}} f(x - t) \frac{dt}{t},$$

work through the chapter's application of Cotlar's lemma.

- (a) Prove $|T_j f(x)| \leq 4M f(x)$.
- (b) Write T_j as convolution with the dyadic kernel K_j .
- (c) For $i < j$, prove the kernel estimate $\|K_i * K_j\|_1 \lesssim 2^{-(j-i)}$.
- (d) Deduce a uniform L^2 bound for finite sums $\sum_{j=n}^m T_j$.



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Exercise E13

Example 9.2: dyadic Hilbert-transform truncations

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Exercise E13

Example 9.2: dyadic Hilbert-transform truncations

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EXERCISE E14 CORE

Carleson measures and tent regions

SOURCE LOCUS: Definition 9.3 and Lemma 9.4, Chapter 9, §2, pp. 197–198

CORE

A positive measure ν on $\mathbb{R}_+^{n+1}$ is Carleson if

$$\nu(Q \times (0, \ell(Q))) \leq C|Q|$$

for every cube $Q \subset \mathbb{R}^n$.

- (a) Define the tent region $\widehat{E}$ over an open set E .
- (b) Reconstruct the dyadic decomposition proof of

$$\nu(\widehat{E}) \leq C\|\nu\| |E|.$$

- (c) Explain the role of the cube whose side length is twice that of a maximal dyadic cube.
- (d) Show how a ball-based formulation compares with the cube formulation.



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Exercise E14

Carleson measures and tent regions

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Exercise E14

Carleson measures and tent regions

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EXERCISE E15 ADVANCED

Characterizing Carleson measures through convolution

ADVANCED

SOURCE LOCUS: Theorem 9.5, Chapter 9, §2, pp. 198–199

Let ϕ be bounded, integrable, positive, radial, and decreasing, and put $\phi_t(x) = t^{-n}\phi(x/t)$. Reconstruct the equivalence between the Carleson condition and

$$\int_{\mathbb{R}_+^{n+1}} |\phi_t * f(x)|^p d\nu(x, t) \leq C \int_{\mathbb{R}^n} |f(x)|^p dx.$$

- (a) Prove the forward direction using the distribution-function identity.
- (b) Dominate the superlevel set by the tent over the Hardy–Littlewood maximal level set.
- (c) Prove the converse by testing on a ball characteristic function.



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Exercise E15

Characterizing Carleson measures through convolution

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Characterizing Carleson measures through convolution

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Exercise E15

Characterizing Carleson measures through convolution

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**EXERCISE E16 ADVANCED****BMO produces a Carleson measure**

SOURCE LOCUS: Theorem 9.6, Chapter 9, §2, pp. 199–200

ADVANCED

Let $b \in BMO$ and $\psi \in \mathcal{S}$ satisfy $\int \psi = 0$. Define

$$d\nu(x, t) = |b * \psi_t(x)|^2 \frac{dx dt}{t}.$$

- (a) Decompose $b - b_{Q^*}$ into the local and far pieces used in the chapter.
- (b) Estimate the local term using the BMO square-function characterization.
- (c) Use the rapid decay of ψ and the dyadic annuli Q_k to control the far term.
- (d) Conclude $\|\nu\| \lesssim \|b\|_*^2$.



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Exercise E16

BMO produces a Carleson measure

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Exercise E16

BMO produces a Carleson measure

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Exercise E16

BMO produces a Carleson measure

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EXERCISE E17 CORE

The weak boundedness property

SOURCE LOCUS: Definition 9.8, Chapter 9, §3, pp. 201–202

CORE

Let $B \subset C_c^\infty(\mathbb{R}^n)$ be bounded. The weak boundedness property asks for a constant C_B such that

$$|(T\phi_{1,x,R}, \phi_{2,x,R})| \leq C_B R^n$$

for all $\phi_1, \phi_2 \in B$, $x \in \mathbb{R}^n$, $R > 0$.

- (a) Verify the scaling of the normalized test functions.
- (b) Explain why an L^2 -bounded operator automatically has WBP.
- (c) Reconstruct the anti-symmetric-kernel argument showing that the operator defined by the principal-value bilinear form has WBP.
- (d) Explain why differentiation has the zero kernel but fails WBP.



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Exercise E17

The weak boundedness property

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Exercise E17

The weak boundedness property

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**PROBLEM P05 ADVANCED****The three conditions of the $T1$ theorem**

SOURCE LOCUS: Theorem 9.9, Chapter 9, §3, pp. 203–204

ADVANCED

Let $T : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$ be associated with a standard kernel.

- (a) Explain why the chapter defines $T1$ and T^*1 through test functions of zero integral.
- (b) State the three conditions: $T1 \in BMO$, $T^*1 \in BMO$, and WBP.
- (c) Explain necessity of each condition from the preceding chapters.
- (d) Explain why the theorem is genuinely stronger than a mere kernel estimate.



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Problem P05

The three conditions of the T_1 theorem

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Problem P05

The three conditions of the T_1 theorem

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Problem P05

The three conditions of the T_1 theorem

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EXERCISE E18 CORE

The anti-symmetric reduction

SOURCE LOCUS: Corollary 9.10, Chapter 9, §3, p. 203

CORE

Assume the standard kernel is anti-symmetric and T is defined by the principal-value bilinear form.

- (a) Prove that T has the WBP.
- (b) Show that $T^*1 = -T1$.
- (c) Deduce from the $T1$ theorem that L^2 boundedness is equivalent to $T1 \in BMO$.



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Exercise E18

The anti-symmetric reduction

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Exercise E18

The anti-symmetric reduction

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EXERCISE E19 ADVANCED

Example 9.11 and the Calderón commutators

SOURCE LOCUS: Example 9.11 and Corollary 9.12, Chapter 9, §3, pp. 203–204

ADVANCED

Let $A : \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz and define

$$K_k(x, y) = \frac{(A(x) - A(y))^k}{(x - y)^{k+1}}, \quad k \in \mathbb{N}.$$

- (a) Verify the standard-kernel estimates for K_k in terms of $\|A'\|_\infty$.
- (b) Use the $T1$ theorem in the anti-symmetric case to reduce the problem to controlling $T_k 1$ in BMO.
- (c) Reconstruct the induction suggested in the chapter and identify where integration by parts enters.
- (d) State the final L^2 estimate in the form $\|T_k\|_{2 \rightarrow 2} \lesssim C^k \|A'\|_\infty$.



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Exercise E19

Example 9.11 and the Calderón commutators

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Example 9.11 and the Calderón commutators

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Exercise E19

Example 9.11 and the Calderón commutators

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EXERCISE E20 ADVANCED

The Cauchy kernel along a Lipschitz curve

SOURCE LOCUS: Corollary 9.13, Chapter 9, §3–4, p. 205

ADVANCED

Consider

$$K(x, y) = \frac{1}{x - y + i(A(x) - A(y))}.$$

Assume $\|A'\|_\infty < 1$.

- (a) Expand $K(x, y)$ as a convergent series in Calderón commutator kernels.
- (b) Identify the smallness condition needed for the expansion.
- (c) Use the commutator bounds to deduce that the associated operator is Calderón–Zygmund.



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Exercise E20

The Cauchy kernel along a Lipschitz curve

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Exercise E20

The Cauchy kernel along a Lipschitz curve

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**PROBLEM P06 CHALLENGE****The zero- $T1$ proof architecture**

SOURCE LOCUS: Lemmas 9.14–9.16, Chapter 9, §4, pp. 205–210

CHALLENGE

Assume first that $T1 = T^*1 = 0$ and T has WBP. Follow the chapter's actual proof architecture.

- (a) Introduce ϕ_j , S_j and Δ_j and write the finite decomposition

$$R_N = \sum_{j=-N}^N (S_j T \Delta_j + \Delta_j T S_j - \Delta_j T \Delta_j).$$

- (b) Prove $R_N \rightarrow T$ in the distributional sense using the WBP.
- (c) Derive the kernel bounds for the dyadic pieces $T_j = S_j T \Delta_j$.
- (d) Prove the almost-orthogonality estimate

$$\|T_j T_k^*\|_{2 \rightarrow 2} + \|T_j^* T_k\|_{2 \rightarrow 2} \lesssim 2^{-\delta|j-k|}.$$

- (e) Apply Cotlar's lemma to conclude uniform L^2 boundedness.



PRINCETON PROOF SHEET

Problem P06

The zero- T_1 proof architecture

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Problem P06

The zero- T_1 proof architecture

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Problem P06

The zero- T_1 proof architecture

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Problem P06

The zero- T_1 proof architecture

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PROBLEM P07 CHALLENGE

Paraproducts and the general $T1$ theorem

CHALLENGE

SOURCE LOCUS: Lemma 9.17, Chapter 9, §4, pp. 210–212

Given $b \in BMO$, reconstruct the chapter's paraproduct construction of a Calderón–Zygmund operator L satisfying

$$L1 = b, \quad L^*1 = 0.$$

Use radial functions $\phi, \psi \in \mathcal{S}$ with $\int \phi = 1$ and $\int \psi = 0$.

- (a) Derive the kernel formula of the paraproduct.
- (b) Prove its size and first-derivative bounds in the standard-kernel form.
- (c) Prove L^2 boundedness using the Carleson measure characterization from Chapter 9.
- (d) Verify $L^*1 = 0$ from the cancellation of ψ .
- (e) Explain the choice of normalization constant c through

$$c \int_0^\infty |\widehat{\psi}(t\xi)|^2 \frac{dt}{t} = 1, \quad \xi \neq 0.$$



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Problem P07

Paraproducts and the general T_1 theorem

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Problem P07

Paraproducts and the general T_1 theorem

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Problem P07

Paraproducts and the general T_1 theorem

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Problem P07

Paraproducts and the general T_1 theorem

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PROBLEM P08 CHALLENGE

Reconstruct the full David–Journé reduction

CHALLENGE

SOURCE LOCUS: Theorem 9.9 and Lemma 9.17, Chapter 9, §§3–4, pp. 203–212

Build a complete proof map of the $T1$ theorem without hiding the main structural reductions.

- (a) Start with arbitrary T satisfying the three $T1$ hypotheses and set $b_1 = T1$, $b_2 = T^*1$.
- (b) Use Lemma 9.17 to construct Calderón–Zygmund operators L_1, L_2 with prescribed $T1$ and T^*1 data.
- (c) Form $\tilde{T} = T - L_1 - L_2^*$ and verify $\tilde{T}1 = \tilde{T}^*1 = 0$.
- (d) Apply the zero- $T1$ case and reconstruct the final L^2 estimate.
- (e) End with a one-page conceptual diagram explaining how BMO, Carleson measure, WBP, paraproducts, and almost orthogonality fit together.



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Problem P08

Reconstruct the full David-Journ´e reduction

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PRINCETON PROOF SHEET

Problem P08

Reconstruct the full David-Journ´e reduction

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PRINCETON PROOF SHEET

Problem P08

Reconstruct the full David-Journ'e reduction

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Problem P08

Reconstruct the full David-Journ´e reduction

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Synthesis Page

RESEARCHER'S MARGIN

Chapters VIII–IX: the structural picture

Write, in your own notation, the shortest route you can find from

dyadic frequency decomposition $\longrightarrow$ square-function control $\longrightarrow$ multiplier theory $\longrightarrow$ singular integrals on curved sets

and then from

Carleson measure $\longrightarrow$ BMO $\longrightarrow$ WBP + $T1, T^*1 \longrightarrow L^2$ boundedness.



DESIGN NOTE. The visual system intentionally combines the supplied workbook’s warm paper, orange rule, large mathematical type, spacious proof sheets, and restrained architectural imagery with an abstract Fuld Hall motif from official Institute for Advanced Study media. The workbook is a private educational artifact and is not an official publication of Princeton University or the Institute for Advanced Study.